

Notes: Gentzkow (QJE, 2006)

Television and Voter Turnout

Contents

1	Big picture	3
2	Data and measurement (institutional context and key objects)	4
2.1	Why this setting is useful	4
2.2	Television availability data	4
2.3	Geographic units and DMAs	4
2.4	Election-return data and turnout measure	5
2.5	National Election Study data	5
2.6	Newspaper and radio data	5
2.7	Descriptive facts about diffusion and turnout	6
2.8	Who got television first?	6
3	Models and empirical specifications (all equations + interpretation)	6
3.1	Baseline fixed-effects specification	6
3.2	How television exposure is coded	7
3.3	Election competitiveness control	7
3.4	Relative-trends specification	7
3.5	Time-polynomial interactions with observables	8
3.6	Subsample design within demographically similar groups	8
3.7	Presidential versus off-year specifications	8
3.8	Heterogeneity and nonlinear effects	8
3.9	Mechanism regressions in the 1952 NES	9
3.10	District-fragmentation interaction	9
4	Identification and empirical design (what makes the estimates credible)	10
4.1	Why timing variation is useful	10
4.2	Why within-DMA heterogeneity matters	10
4.3	Placebo and orthogonality tests	10
4.4	Why demographics are such a central threat	11
4.5	Why off-year elections are especially informative	11
4.6	What the paper can and cannot identify	11
5	Tables and figures	11

5.1	Figures I–V and Table I: diffusion is fast, staggered, and not random	12
5.2	Table II: the leftover timing variation looks close to idiosyncratic	12
5.3	Figure VII and Tables III–IV: turnout falls after television arrives	13
5.4	Table V and Table VI: the effect survives within-group comparisons and is larger where diffusion and sensitivity are stronger	14
5.5	Figure VIII and Table VII: television crowds out newspapers and radio	15
5.6	Table VIII: television reduces political knowledge, especially for congressional races	16
5.7	Table IX: district fragmentation amplifies the off-year turnout effect	16
6	Short summary	17
7	Conclusion	17
7.1	Core conclusion	17
7.2	Economic intuition	17
7.3	Takeaway	18

1 Big picture

- **Question.** Did the arrival of television increase democratic participation by informing citizens better, or did it instead reduce turnout by crowding out other media and diverting attention toward entertainment?
- **Setting.** The paper studies the staggered introduction of television across U.S. media markets from the 1940s through the early 1950s, using county-level election data and survey evidence on information and media use.
- **Core challenge.** Large and rich places got television earlier, so simple cross-sectional comparisons would confound television with urbanization, income, education, and preexisting political trends.
- **Main conceptual contribution.** The paper turns television's diffusion into a differences-in-differences design. It exploits the fact that television spread at very different times across markets, that World War II and the FCC licensing freeze injected quasi-exogenous timing variation, and that stations broadcast to heterogeneous counties within the same market.
- **Main mechanism hypothesis.** Television improved entertainment much more than it improved political information. Because newspapers and radio provided more political coverage, especially for local and congressional races, television likely reduced turnout by crowding out those media and lowering political knowledge.
- **Main empirical design.** The baseline specification regresses county turnout on *years since television arrived* with county fixed effects and census-region-by-year fixed effects, then adds rich demographic controls and interactions of demographics with flexible time trends.
- **Main baseline result.** In the preferred turnout specification, each additional year of television lowers congressional turnout by about **0.136 percentage points**. When presidential and nonpresidential years are separated, the effect is about **-0.196** percentage points per year in off-year congressional elections and about **-0.067** in presidential years.
- **Key heterogeneity result.** Television matters much more in **off-year congressional elections** than in presidential years. This matches the mechanism: local and congressional races receive much more newspaper coverage and much less television coverage than presidential elections.
- **Key mechanism evidence.** In the 1952 National Election Study, access to television reduces the probability of getting campaign information from **newspapers by 11.4 percentage points** and from **radio by 28.3 percentage points**, while increasing the probability of getting campaign information from television by **44.5 percentage points**. It also reduces the ability to name congressional candidates.
- **Main credibility result.** The turnout effect survives county fixed effects, region-year fixed effects, controls for changing demographics, fourth-order time-trend interactions with base-year observables, placebo tests, and estimations run within demographically similar subsets of counties.

- **Magnitude.** Gentzkow concludes that television can account for between roughly **one quarter and one half** of the post-1950 decline in turnout, with the strongest effect in off-year elections.
- **Economic takeaway.** New media can reduce political participation even when they lower the cost of information, if they lower the cost of entertainment even more and crowd out existing information-rich media.

2 Data and measurement (institutional context and key objects)

2.1 Why this setting is useful

- Television diffused **gradually but rapidly enough** that its arrival created large time variation in actual exposure.
- Some counties got television more than **ten years** after the earliest markets.
- World War II and the **FCC freeze from 1948 to 1952** disrupted the smooth rollout of stations, creating extra quasi-exogenous timing variation.
- A station's broadcast area covered a broad set of counties, including both urban and rural places, so counties with similar demographics could differ in television timing because they lay inside or outside the reach of a nearby market.

Interpretation. This is unusually favorable historical variation: adoption is large, staggered, and tied to regulatory and wartime shocks rather than only to household tastes.

2.2 Television availability data

- Television availability comes from *Television Factbook* issues from **1948–1970**.
- For each station, the source lists its location, signal strength, ownership, network affiliation, and start date.
- A county is coded as receiving television in the first year a station in its market broadcast for at least **four months**.

Interpretation. The paper does not proxy television with later ownership or current channel count; it measures the actual timing of first access.

2.3 Geographic units and DMAs

- Markets are defined using **Designated Market Areas (DMAs)**, the standard television-market geography.
- Counties are assigned to the DMA where they devote most viewing hours.

- Although DMA definitions are based on modern viewing patterns, the paper argues that historical station coverage was similar because FCC regulation limited overlap and interference.
- Counties in very late-entry DMAs (1960 or later) are dropped because they likely already received spillover signals from neighboring markets. These 13 DMAs account for only about **0.7%** of the 1950 population.

Interpretation. The treatment is effectively market-level first access, mapped into counties.

2.4 Election-return data and turnout measure

- County-level election returns come from ICPSR’s historical election data.
- The turnout denominator is the **legally eligible electorate**, not registered voters.
- Counties are kept if turnout data are available for a majority of the years from **1940 to 1972**, leaving **3081 counties**.
- The main dependent variable is **votes cast for Congress as a share of the eligible electorate**.

Interpretation. Congressional turnout is the main outcome because the mechanism predicts especially large effects in the relatively local elections that television covered poorly.

2.5 National Election Study data

- The paper uses the **1952 National Election Study**, a nationally representative survey of **1899 voting-age citizens** interviewed before and after the 1952 election.
- 1952 is ideal because it was the end of the FCC freeze, so idiosyncratic variation in access to television was especially large.
- The survey contains information on media use, political knowledge, voting behavior, demographics, and county of residence.

Interpretation. The survey lets the paper move beyond turnout and test the information-crowding-out mechanism directly.

2.6 Newspaper and radio data

- State-level newspaper circulation comes from *Editor and Publisher Yearbook* for **1946–1971**.
- The paper converts total daily circulation into circulation per capita using census population data.
- For radio, the paper uses industry evidence and survey responses rather than a county-by-year panel analogous to the television data.

Interpretation. The media-substitution evidence combines a historical circulation panel with survey-based measures of campaign information sources.

2.7 Descriptive facts about diffusion and turnout

- Television was licensed for commercial broadcasting on **July 1, 1941**.
- By **1960**, **87%** of American households had a television set and watched about **5.5 hours per day**.
- In the largest counties, about **20%** of households had television by 1950 and about **80%** by 1955.
- Television households were already watching more than **4.5 hours per day** by 1950.
- Turnout declines sharply after 1960, especially outside the South where remobilization of Black voters does not mask the fall.

Interpretation. Television’s diffusion was fast enough, and turnout’s decline large enough, that the treatment can plausibly move the outcome within a relatively short empirical window.

2.8 Who got television first?

- Counties with earlier television were larger, denser, more urban, richer, older, and had higher baseline turnout.
- Table I shows especially large differences between counties with pre-1945 television and later groups.
- This is exactly why cross-sectional identification is not enough and why the paper invests heavily in controls and within-group comparisons.

Interpretation. Early television markets are observably different, so credible identification must come from changes within counties over time and from comparisons among similar counties.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline fixed-effects specification

The main empirical model is

$$Y_{it} = \alpha_i + \delta_{rt} + \gamma TV_{it} + \beta X_{it} + \varepsilon_{it}, \quad (1)$$

where i indexes counties, t indexes years, and r indexes census regions.

- Y_{it} is voter turnout or another outcome of interest.
- α_i are county fixed effects.
- δ_{rt} are region-by-year fixed effects.
- TV_{it} measures the scale of television exposure.
- X_{it} includes competitiveness and a set of county controls.

Interpretation. The identifying variation is whether turnout in a county changes differentially after television arrives, relative to counties in the same broad region-year environment.

3.2 How television exposure is coded

The baseline treatment grows linearly after television is introduced:

$$TV_{it} = \mathbf{1}(t \geq \tau_i)(t - \tau_i), \quad (2)$$

where τ_i is the first year television arrives in county i .

- Counties with stations before World War II are assigned a first-television year of **1946**, because prewar penetration was negligible.
- The specification allows the effect of television to accumulate over time rather than showing up as a one-time jump.

Interpretation. Gentzkow views television as a gradually intensifying treatment because ownership diffused, programming improved, and informational effects likely accumulated slowly.

3.3 Election competitiveness control

All turnout regressions include the absolute difference between the Republican and Democratic vote shares in the election, plus a missing-value dummy when needed.

Interpretation. Closer elections raise turnout, so the paper conditions on competitiveness to avoid attributing ordinary electoral incentives to television.

3.4 Relative-trends specification

Before moving to the full years-of-television design, the paper compares three groups of counties:

- counties with television before 1945,
- counties first getting television in 1945–1951,
- counties first getting television in 1952 or later.

It estimates differential pre- and post-1952 trends for the first two groups relative to the last.

Interpretation. This is a transparent event-study-style precursor: early-TV counties should show a relative decline before 1952, and that relative decline should flatten once the late-TV counties also get access.

3.5 Time-polynomial interactions with observables

To absorb smooth differential trends related to market size and development, the paper adds interactions between a fourth-order polynomial in time and base-year observables:

$$TimePoly_t \times \{\log Population, \log Income, Turnout_{1940}, \text{missing } Turnout_{1940}\}. \quad (3)$$

In additional robustness checks it also interacts time with percent urban, population density, and percent non-White.

Interpretation. This says: even if large, rich, early-TV places were already on different trajectories, a flexible function of those trajectories is partialled out.

3.6 Subsample design within demographically similar groups

The paper also partitions counties into thirds by characteristics such as population, density, urbanization, education, income, and baseline turnout, then reruns the turnout regressions within each third.

Interpretation. This is the most literal version of the paper’s identifying thought experiment: compare similar counties, and let treatment timing be driven by broadcast geography rather than underlying social change.

3.7 Presidential versus off-year specifications

The paper allows separate treatment effects in presidential and nonpresidential years:

$$Y_{it} = \alpha_i + \delta_{rt} + \gamma_1(TV_{it} \times OffYear_t) + \gamma_2(TV_{it} \times PresYear_t) + \beta X_{it} + \varepsilon_{it}. \quad (4)$$

Interpretation. This is central to the mechanism. If television mainly crowds out local political information, the effect should be stronger when the most salient races are congressional, gubernatorial, or local rather than presidential.

3.8 Heterogeneity and nonlinear effects

The paper estimates richer models including:

$$Y_{it} = \alpha_i + \delta_{rt} + \gamma_1 TV_{it} + \gamma_2 TV_{it}^2 + \sum_k \theta_k(TV_{it} \times Z_{ik}) + \beta X_{it} + \varepsilon_{it}, \quad (5)$$

where Z_{ik} includes demographics, predicted television diffusion, predicted turnout sensitivity, and the year television first arrived.

- **Predicted diffusion** comes from a regression of 1960 television penetration on demographics and first-TV-year dummies.
- **Predicted turnout sensitivity** comes from a regression allowing turnout's response to election closeness to vary by demographics.

Interpretation. These interactions ask where television should bite most: where it spread fastest and where there were more marginal voters.

3.9 Mechanism regressions in the 1952 NES

For survey outcomes, the paper runs cross-sectional probit models of the form

$$Pr(Outcome_i = 1) = \Phi(\alpha + \gamma TV_i + \beta X_i), \quad (6)$$

where TV_i is a dummy for whether the respondent's county had television by November 1952.

Outcomes include:

- whether the respondent got campaign information from newspapers,
- from radio,
- from television,
- and whether the respondent could correctly name candidates in House, Senate, and gubernatorial races.

Interpretation. These are reduced-form first-stage and mechanism tests: did television change where people got campaign information, and did that change lower concrete political knowledge?

3.10 District-fragmentation interaction

To sharpen the local-information mechanism, the paper estimates turnout regressions that interact television with the number of congressional districts in the DMA:

$$Y_{it} = \alpha_i + \delta_{rt} + \gamma_1 TV_{it} + \gamma_2 (TV_{it} \times Districts_i) + \beta X_{it} + \varepsilon_{it}. \quad (7)$$

Interpretation. If a market is split across many congressional districts, television should find it harder to cover any one local race well. The turnout effect should therefore be more negative in fragmented markets, especially in off-year elections.

4 Identification and empirical design (what makes the estimates credible)

4.1 Why timing variation is useful

- The key source of identification is the staggered rollout of television across markets.
- Two shocks made this timing less smooth and more plausibly exogenous: **World War II** and the **1948–1952 FCC freeze**.
- Figure V shows three obvious waves of introduction: prewar, postwar but prefreeze, and post-freeze.

Interpretation. Timing was not random, but it was not a smooth market-size ranking either. The shocks add identifying discontinuities that strengthen the panel design.

4.2 Why within-DMA heterogeneity matters

- Even though the earliest television cities were big and wealthy, their signals reached nearby counties that were often much smaller and more rural.
- Gentzkow's Chicago-DMA example contrasts highly urban Cook County with rural Newton County, Indiana, both exposed to Chicago television.

Interpretation. This geographic feature makes it plausible to compare counties with similar observables but different access timing driven by proximity to a large city's broadcast radius.

4.3 Placebo and orthogonality tests

- Table II studies whether, after partialing out log population, log income, 1940 turnout, and region effects, the remaining variation in the timing of television is still correlated with county observables.
- In the full-sample residualized regression, the remaining demographics are neither jointly nor individually significant: the reported F -statistic is about **1.04**, with **0 of 5** coefficients individually significant.
- In the 1952 NES sample, the analogous placebo also shows little systematic correlation: the F -statistic is about **1.31**, with just **1 of 28** controls individually significant.

Interpretation. The paper cannot prove random assignment, but after residualizing the main determinants of entry timing, the leftover variation looks close to idiosyncratic.

4.4 Why demographics are such a central threat

- Counties that got television early were larger, richer, more urban, and higher-turnout even before treatment.
- Any independent social trend beginning in big cities and diffusing outward could mimic a television effect.
- This is exactly why the paper adds flexible trend interactions and also checks the estimates within homogeneous subsets of counties.

Interpretation. The paper’s main identifying burden is to separate television from broader mid-century urban social change.

4.5 Why off-year elections are especially informative

- Off-year congressional elections receive much richer coverage in newspapers than on television.
- Presidential elections receive more national television coverage.
- Therefore, an information-crowding-out mechanism predicts larger turnout and knowledge effects in off-year elections.

Interpretation. The off-year versus presidential comparison is a built-in mechanism test, not just a robustness exercise.

4.6 What the paper can and cannot identify

- The paper credibly identifies a reduced-form effect of television’s arrival, not a structural model of exactly how households allocate time across all information and entertainment choices.
- The survey evidence is cross-sectional, so it is less clean than the county panel.
- Newspaper circulation is only available at the state level, so those results are less granular than the turnout results.
- The long-run magnitude depends on how nonlinear the effect is over time, and the paper shows that those nonlinear terms are hard to pin down precisely.

Interpretation. The most convincing contributions are the medium-run county panel effects and the fact that several very different pieces of evidence line up with the same story.

5 Tables and figures

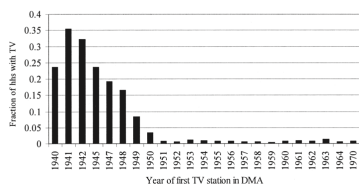


FIGURE I
1950 Television Penetration by First Television Year
Note: Years in which no county received their first station are omitted from the figure.

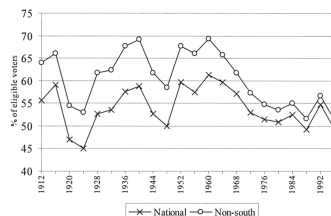


FIGURE II
Congressional Turnout in Presidential Election Years

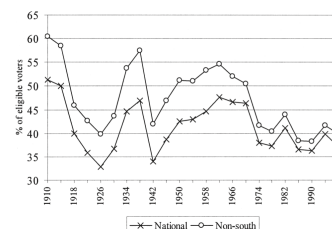


FIGURE III
Congressional Turnout in Nonpresidential Years

5.1 Figures I–V and Table I: diffusion is fast, staggered, and not random

Read.

- Figure I shows that counties recorded as having television before 1950 already had substantial 1950 ownership, while post-1950 counties had almost none.
- Figures II and III show the long-run decline in congressional turnout, especially after 1960.
- Figure IV shows rapid diffusion: in the largest counties, television penetration rises from about 20% in 1950 to around 80% by 1955.
- Figure V shows the three waves of market entry created by the prewar rollout, wartime interruption, and FCC freeze.
- Table I shows that early-TV counties are much larger, denser, richer, more urban, and higher-turnout than later-TV counties.

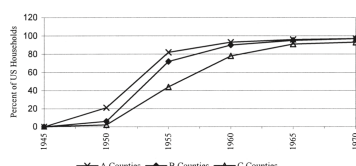


FIGURE IV
Percent of U. S. Households with Television by County Size
Note: Data are from Nielsen Television Index as quoted in Sterling [1984]. "A Counties" are all counties in the 25 largest metropolitan areas. "B Counties" are all counties not in A with populations of over 150,000 or in metropolitan areas.

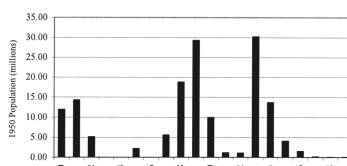


FIGURE V
1950 Population of Counties by Year Television Was Introduced

	TABLE I SUMMARY STATISTICS				
	Mean by year of first TV station				
	Mean	St. dev.	1940-1944	1945-1951	1952 and later
Percent urban (1950)	29.0	27.1	60.4	31.6	26.0
Population (1960 '000)	59,428	208,549	466,960	76,057	31,278
Income per capita (1959)	\$1,352	405	\$2,044	\$1,391	\$1,298
% high school (1950)	27.2	10.9	35.9	27.0	26.8
Median age (1950)	28.5	3.92	32.8	28.7	28.1
Percent non-White (1950)	10.9	17.0	4.48	10.1	11.7
Persons per square mile (1950)	209	2,077	3,006	235	67.5
Base year turnout (1940)	56.5	23.6	73.1	57.4	55.2
N	2958		83	1052	1823

Percent high school is the fraction of the population aged 25 and older who completed high school. Income per capita is measured in 1959 dollars.

Economic message. The raw data justify the design and the caution. Television access changed fast enough to matter, but early adopters were systematically different, so identification must come from panel methods and rich controls.

5.2 Table II: the leftover timing variation looks close to idiosyncratic

Read.

- Placebo regressions ask whether, after partialing out the main determinants of timing, the residual variation in postfreeze entry still lines up with observables.

TABLE II
PLACEBO REGRESSIONS WITH POSTFREEZE DUMMY AS DEPENDENT VARIABLE
(*F*-test for joint significance of demographics)

	<i>F</i>	# Significant coefficients
Lowest third of counties by:		
Population (1960 '000)	1.32	0 of 8
Population per square mile (1950)	2.23*	1 of 8
Percent urban (1950)	4.38**	1 of 8
Income per capita (1959)	1.50	0 of 8
% high school (1950)	2.29*	1 of 8
1940 turnout	1.92	1 of 8
Full sample after controlling for log population, log income, and 1940 turnout	1.04	0 of 5
National Election Study sample	1.31	1 of 28

* Significant at the 5 percent level.

** Significant at the 1 percent level.

The two columns show *F*-statistics for the joint significance of the demographic variables in each regression and the number of coefficients significant at the 5 percent level, respectively. The "lowest third of counties" regressions use only counties falling into the lowest third of the indicated demographics; the dependent variable is a dummy equal to one if television was introduced in 1952 or later and the right-hand-side variables are log total income, log population, percent urban, percent non-White, percent high school, median age, population density, and 1940 turnout; region dummies are also included but not reported in the *F*-test. In the "full sample" regression, the dependent variable is the residual from a regression of the postfreeze dummy on log population, log total income, 1940 turnout, and region dummies; the right-hand variables are percent urban, percent non-White, percent high school, median age, and population density. In the "National Election Study" regression, the dependent variable is the residual from a regression of a dummy for having TV in 1952 on DMA log population and log total income, county percent non-White, and region dummies; the right-hand side variables are individual-level dummies for male, White, age categories, education categories, occupation categories, income in the highest category, and missing values, as well as number of children and log income.

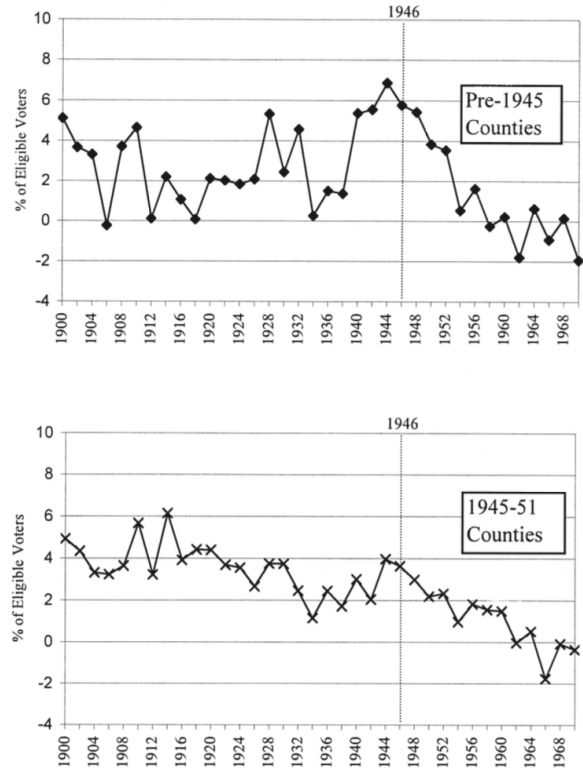


FIGURE VII
Voter Turnout by Year of Television's Introduction
(Relative to Post-1951 Counties)

- In the full sample after residualizing on log population, log income, and 1940 turnout, the remaining demographics are not jointly significant.
- The same is true in the 1952 NES sample.

Economic message. The paper cannot claim random timing, but its residual identifying variation is not obviously driven by observable county or household characteristics.

5.3 Figure VII and Tables III–IV: turnout falls after television arrives

Read.

- Figure VII and Table III show that counties getting television earlier experience a negative relative turnout trend before the later counties get it, and that the relative decline flattens once the later counties also receive television.
- Table IV provides the core estimates. In the specification with full controls, the overall effect is about **-0.136** percentage points of turnout per year of television.
- When presidential and off-year elections are separated, the preferred estimates are roughly **-0.196** for off-year elections and **-0.067** for presidential years.
- Adding further time-polynomial interactions with percent urban, population density, and per-

TABLE III
DIFFERENTIAL TRENDS IN VOTER TURNOUT BY YEAR OF TELEVISION'S INTRODUCTION
(Relative to counties where television was introduced in 1952 or later)

	(1)	(2)	(3)
<i>First year of TV before 1945</i>			
Relative trend, 1944–1950	-.513 (.0775)	-.414 (.0777)	-.399 (.0750)
Relative trend, 1952+	-.306 (.0618)	-.006 (.0542)	.026 (.0535)
<i>First year of TV 1945–1951</i>			
Relative trend, 1946–1950	-.125 (.0391)	-.101 (.0396)	-.106 (.0386)
Relative trend, 1952+	-.139 (.0241)	-.030 (.0192)	-.0219 (.0194)
<i>Demographics</i>			
<i>Time polynomial interactions</i>		X	X
<i>County, year-region dummies:</i>	X	X	X
R^2	.914	.925	.927
N	46003	46003	46003

Standard errors are clustered by county. Dependent variable is the percentage of legally eligible voters casting a vote for Congress. All trends are relative to counties getting television in 1952 or later. When years are given for a relative trend (e.g., 1944–1950), the coefficient is on a variable equal to zero in the first year (1944), one in the second year (1945), and so forth. All regressions include county fixed effects, separate year fixed effects by census region, the absolute difference between the Democrat and Republican vote percentages, and a dummy for a missing value for the absolute difference. Demographics are log population, population density, percent urban, percent non-White, median income, median age, and percent with high-school education. Time polynomial interactions are a fourth-order polynomial in time interacted with 1960 log population, 1959 log total income, 1940 congressional turnout, and an indicator for a missing value of 1940 congressional turnout. Note that the levels of these variables are absorbed by the county fixed effects.

TABLE IV
REGRESSIONS OF TURNOUT ON YEARS OF TELEVISION

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Years of TV	-.416 (.0486)	-.136 (.0412)					
Years of TV * nonpresidential year			-.489 (.0577)	-.196 (.0478)	-.193 (.0483)	-.187 (.0478)	-.188 (.0477)
Years of TV * presidential year			-.332 (.0468)	-.067 (.0438)	-.067 (.0443)	-.059 (.0437)	-.056 (.0434)
<i>Additional time-polynomial interactions:</i>							
<i>Time * percent urban</i>					X		
<i>Time * population density</i>						X	
<i>Time * percent non-White</i>							X
<i>Demographics</i>		X		X	X	X	X
<i>Time polynomial interactions</i>		X		X	X	X	X
<i>County, year-region dummies:</i>	X	X	X	X	X	X	X
R^2	.913	.927	.913	.927	.927	.927	.927
N	46003	46003	46003	46003	46003	46003	46003

Standard errors are clustered by county. Years of TV is the number of years since the first year in which a commercial station was broadcasting in the county for at least three months. The dependent variable is the percentage of legally eligible voters casting votes for congress. Fixed effects, demographics, and time polynomial interactions are as in Table III. All regressions include the absolute difference between the Democratic and Republican vote percentage. Additional time-polynomial interactions are interactions of a fourth-degree polynomial in time with the indicated demographics.

cent non-White barely changes the result.

Economic message. The main reduced-form result is strong, stable, and much larger where the local-information mechanism predicts it should be.

5.4 Table V and Table VI: the effect survives within-group comparisons and is larger where diffusion and sensitivity are stronger

TABLE V
REGRESSIONS OF TURNOUT ON YEARS OF TELEVISION FOR SUBSETS OF COUNTIES
(Coefficient on years of television)

	Lowest third	Second third	Highest third
<i>Counties partitioned by:</i>			
Population	-.332** (.1045)	-.228** (.0798)	-.443** (.0819)
Population density	-.447** (.1041)	-.239** (.0798)	-.289** (.0827)
Percent urban	-.380** (.0940)	-.296** (.0885)	-.440** (.0733)
Per capita income	-.127 (.0984)	-.228** (.0803)	-.578** (.0817)
% high school	-.275** (.0940)	-.282** (.0798)	-.706** (.0752)
1940 turnout	-.160* (.0778)	-.365** (.0788)	-.518** (.0781)

* Significant at the 5 percent level.

** Significant at the 1 percent level.

Standard errors are clustered by county. The table shows the coefficient on years of TV from regressions of turnout on the absolute difference between the Democrat and Republican vote percentages, county dummies, and year-region dummies. The years of TV variable is as in Table IV. Each column gives the coefficient from regressions using only counties that fell into the given third of the data, and each row specifies the demographic characteristic on which counties were divided.

TABLE VI
INTERACTIONS AND NONLINEAR EFFECTS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Years of TV	-.128 (.0422)	-.123 (.0413)	-.138 (.0421)	-.114 (.0425)	-.199 (.0528)	-.186 (.0489)	-.160 (.0522)	-.192 (.0543)
Years of TV squared						.00235 (.00005)	.0255 (.0125)	.0104 (.0126)
<i>Interaction with:</i>								
Predicted diffusion (mean = 0, stdev = 8.76)		-.0184 (.00367)		-.0168 (.00365)	-.0171 (.00367)			-.0167 (.00374)
Predicted sensitivity (mean = 0, stdev = 1)			-.0828 (.0138)	-.108 (.0147)	-.111 (.0149)			-.110 (.0150)
First TV year (mean = 0, stdev = 2.58)					-.00761 (.00419)		.0466 (.0252)	.01281 (.0256)
Percent urban (1950)								
Log population (1960)								
Log income per capita (1959)								
% high school (1950)								
Median age (1950)								
Percent non-White (1950)								
Population density (1950)								
1940 turnout								
<i>Demographics</i>	X	X	X	X	X	X	X	X
<i>Time polynomial interactions</i>	X	X	X	X	X	X	X	X
<i>County, year-region dummies:</i>	X	X	X	X	X	X	X	X
R^2	.9273	.9270	.9269	.9270	.9271	.9268	.9268	.9271
N	42832	46003	42832	42832	42832	46003	46003	42832

Standard errors are clustered by county. The dependent variable, fixed effects, demographics, and time polynomial interactions are as in Table III. The Years of TV variable is as in Table IV. All regressions include the absolute difference between the Democrat and Republican vote percentages. All interactions are the Years of TV variable times the interaction variable minus its mean. Predicted diffusion and predicted sensitivity are fitted values from regressions of diffusion and turnout sensitivity on demographics as described in the text.

Read.

- Table V reruns the turnout regression within thirds of the county distribution by population, density, urbanization, income, education, and baseline turnout.
- The coefficient on years of television is negative in **all 18** cells and statistically significant in all

but one.

- Table VI shows that the effect is more negative where predicted television diffusion is faster and where predicted turnout sensitivity is higher.
- A one-standard-deviation increase in predicted diffusion more than **doubles** the television effect; a one-standard-deviation increase in predicted turnout sensitivity raises it by about **60%**.
- Nonlinearities over time are harder to estimate precisely; the quadratic term is unstable across specifications.

Economic message. The negative turnout effect is not an artifact of a particular set of big-city counties. It appears broadly, and it is strongest exactly where actual viewing growth and marginal-voter sensitivity are likely to have been greatest.

5.5 Figure VIII and Table VII: television crowds out newspapers and radio

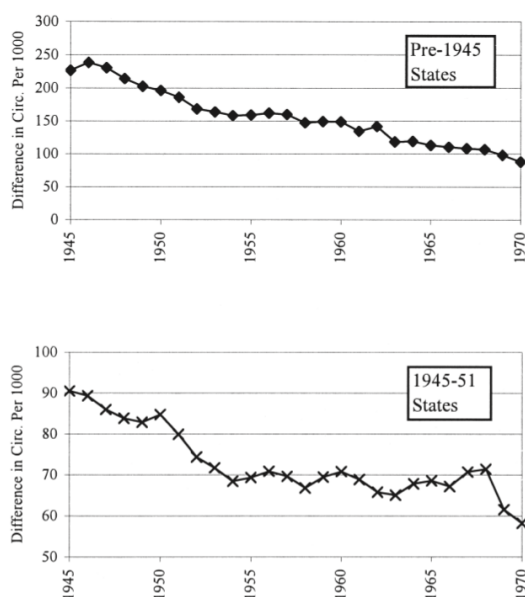


FIGURE VIII
Circulation of Daily Newspapers by Year of Television's Introduction
(Relative to Post-1951 States)

TABLE VII
MEDIA USE IN THE 1952 ELECTION

Dependent variable	Any campaign info from:		
	Newspaper	Radio	Television
TV dummy	−0.114 (.0402)	−0.283 (.0423)	0.445 (.0745)
Log income	0.0752 (.0211)	0.000037 (.0172)	0.1136 (.0339)
Ed: high school	0.119 (.0207)	0.0815 (.0297)	0.1133 (.0312)
Ed: college	0.165 (.0163)	0.0936 (.0515)	0.1224 (.0677)
Mean of dep. var.	.791	.699	.514
R ²	.185	.089	.309
N	1693	1705	1653

Standard errors are clustered by county. Coefficients are marginal effects from probit regressions. In addition to controls shown, all regressions include individual-level dummies for age, occupation, sex, White race, highest income category, political party identification, and missing values of each control, as well as a continuous control for number of children. Each also includes controls for DMA-level log income and population; county-level population, percent urban, population density, percent non-White, median age, median income, and median schooling; and dummies for census regions. The TV dummy is one if the respondent's county had television prior to 1952. The "any campaign info" variables are dummies equal to one if the respondent reported obtaining information from the given source.

Read.

- Figure VIII shows relative declines in newspaper circulation in states that got television earlier.
- Table VII uses the 1952 NES to look directly at campaign information sources.
- Living in a television county reduces the probability of getting campaign information from **newspapers by 11.4 percentage points** and from **radio by 28.3 percentage points**.
- It increases the probability of getting campaign information from **television by 44.5 percentage points**.

Economic message. Television did not simply add another news source. It displaced older media that were richer in political content, especially for local races.

5.6 Table VIII: television reduces political knowledge, especially for congressional races

Read.

- In preelection survey responses, access to television lowers the probability of correctly naming a House candidate by about **24.8 percentage points**; the effects for Senate and governor are small and insignificant.
- In postelection responses, the ability to name the voted-for candidate falls by about **34.0 percentage points** for House races, **23.6 percentage points** for Senate races, and **11.4 percentage points** for gubernatorial races.
- The House effects are the strongest, matching the idea that congressional races were least covered on television.

Economic message. The information channel is not just speculation. Television counties show measurably lower knowledge of who is running, especially in the most local races.

Dep. var.	Able to name candidate preelection:			Able to name candidate postelection:		
	House	Senate	Governor	House	Senate	Governor
TV dummy	−0.248 (.1155)	0.08916 (.1353)	0.0392 (.0811)	−0.340 (.1054)	−0.236 (.0700)	−0.114 (.0700)
Log income	0.0192 (.0373)	0.0344 (.0481)	0.0360 (.0416)	0.0904 (.0371)	0.0742 (.0344)	0.0410 (.0312)
Ed: high school	0.120 (.0553)	0.181 (.0507)	−0.003 (.0291)	0.098 (.0492)	0.0156 (.0409)	0.052 (.0383)
Ed: college	0.198 (.0987)	0.224 (.0690)	0.016 (.0676)	0.227 (.0654)	0.069 (.0814)	0.135 (.0435)
Mean of dep. var.	.261	.280	.272	.489	.662	.755
R ²	.181	.166	.201	.203	.180	.199
N	653	525	612	1011	795	682

Standard errors are clustered by county. Coefficients are marginal effects from probit regressions. TV dummy is equal to one if the respondent's county had a television station in November 1952. Controls are as in Table VII. "Able to name candidate preelection" variables are dummies equal to one if the respondent gave the name of a candidate that they intended to vote for the specified office in the preelection interview, and equal to zero if they simply named a party or mentioned a candidate not running; respondents who said they did not intend to vote for the specified office are omitted. "Able to name candidate postelection" variables are defined analogously for the postelection survey.

	(1)	(2)
<i>(number of districts: mean = 0 stdev = 5.78)</i>		
Off-year elections		
Years of TV	−.374 (.0584)	−.144 (.0486)
Years of TV * number of districts	−.00829 (.00130)	−.00462 (.00126)
Presidential elections		
Years of TV	−.268 (.0479)	−.0597 (.0454)
Years of TV * number of districts	−.0048 (.00139)	−.00119 (.00133)
Demographics		X
Time polynomial interactions		X
County, year * region dummies:	X	X
R ²	.914	.927
N	46003	46003

Standard errors are clustered by county. The dependent variable is the percentage of legally eligible voters casting votes for congress. Number of districts is measured at the DMA-level. All regressions include an interaction between years of television and 1960 DMA population as a control. Years of TV, dependent variable, fixed effects, demographics, and time polynomial interactions are as in Table III. All regressions include the absolute difference between the Democratic and Republican vote percentage.

5.7 Table IX: district fragmentation amplifies the off-year turnout effect

Read.

- Table IX interacts years of television with the number of congressional districts in the DMA.
- In off-year elections, the interaction is significantly negative even after the full set of controls.

- A one-standard-deviation increase in the number of districts raises the magnitude of television’s off-year turnout effect by about **18%**.
- In presidential years, the interaction is small and insignificant once the controls are added.

Economic message. When television has to divide its local political attention across many House races, the turnout effect gets worse exactly where the crowding-out story predicts.

6 Short summary

- Gentzkow exploits staggered television entry across U.S. markets to estimate television’s causal effect on turnout.
- The preferred county-panel estimates show that television reduced congressional turnout, with a much larger effect in off-year elections than in presidential years.
- The evidence is robust to fixed effects, demographic controls, flexible trend interactions, placebo tests, and estimations within similar subsets of counties.
- Mechanism evidence shows that television displaced newspapers and radio as campaign information sources and reduced political knowledge, especially about congressional candidates.
- The paper concludes that television explains a substantial share of the post-1950 decline in turnout because it crowded out information-rich media while expanding entertainment options.

7 Conclusion

7.1 Core conclusion

- Television reduced voter turnout rather than increasing it.
- The effect is strongest where television provided the least useful political information: off-year congressional elections and more locally fragmented media markets.

7.2 Economic intuition

- Television lowered the cost of both entertainment and information.
- But the entertainment margin appears to have dominated: households substituted away from newspapers and radio, consumed less local political information, and became less likely to vote.
- The result is a media-driven reduction in participation rather than a direct ideological persuasion effect.

7.3 Takeaway

- New communication technologies do not automatically deepen democracy.
- A medium can expand overall media consumption while lowering the supply of politically relevant information that citizens actually absorb.
- Gentzkow's paper is a clean reminder that the political effects of media depend on *what is crowded out*, not only on what is newly introduced.